```
find: '/home/lost+found': Permission denied
/home/hadoop/1901
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbte$ cd ~/Desktop
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/Desktop$ cd
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ ls
 1901                  map1.py                    Public
 1902                  map.py                     red1.py
 Desktop               MDriver.java               red.py
 Documents            'MDriver.java\'             snap
 Downloads             MMapper.java               Templates
 eclipse-workspace     MReducer.java              Videos
 hadoop                Music                      wc.jar
 hadoop-3.3.6.tar.gz  'pi$$anonfun$main$1.class'  wordcount
 hadoopdata           'pi$.class'                 Wordcount
 inp.txt               pi.class                   WordCount
 input.txt             Pictures                   WordCount.jar
 i.txt                 pi.scala                   Word.jar
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put 1901 /weather
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put 1902 /weather
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
-rw-r--r--   1 hadoop supergroup     888190 2026-03-13 15:26 /weather/1901
-rw-r--r--   1 hadoop supergroup     888978 2026-03-13 15:26 /weather/1902
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cd ~
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ ls
 1901                  input.txt                  'pi$.class'   wc.jar
 1902                  i.txt                       pi.class     WeatherAnalysis
 Desktop               map1.py                     Pictures     Weather.jar
 Documents             map.py                      pi.scala     wordcount
 Downloads             MDriver.java                Public       Wordcount
 eclipse-workspace    'MDriver.java\'              red1.py      WordCount
 hadoop                MMapper.java                red.py       WordCount.jar
 hadoop-3.3.6.tar.gz   MReducer.java               snap         Word.jar
 hadoopdata            Music                       Templates
 inp.txt              'pi$$anonfun$main$1.class'   Videos
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ jps
7936 NameNode
8740 ResourceManager
10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
8127 DataNode
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
-rw-r--r--   1 hadoop supergroup     888190 2026-03-13 15:26 /weather/1901
-rw-r--r--   1 hadoop supergroup     888978 2026-03-13 15:26 /weather/1902
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
Exception in thread "main" java.lang.UnsupportedClassVersionError: avgtemp/AverageDriver has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version of the Java
Runtime only recognizes class file versions up to 55.0
        at java.base/java.lang.ClassLoader.defineClass1(Native Method)
        at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
        at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
        at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
        at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)
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10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
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hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
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hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
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        at java.base/java.lang.ClassLoader.defineClass1(Native Method)
        at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
        at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
        at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
        at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)
        at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:452)
        at java.base/java.security.AccessController.doPrivileged(Native Method)
        at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:451)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
        at java.base/java.lang.Class.forName0(Native Method)
        at java.base/java.lang.Class.forName(Class.java:398)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
2026-03-13 15:58:55,743 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-13 15:58:55,936 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-13 15:58:55,961 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1773395565039_0001
2026-03-13 15:58:56,117 INFO input.FileInputFormat: Total input files to process : 2
2026-03-13 15:58:56,202 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 15:58:56,319 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773395565039_0001
2026-03-13 15:58:56,319 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 15:58:56,398 INFO conf.Configuration: resource-types.xml not found
2026-03-13 15:58:56,399 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-13 15:58:56,503 INFO impl.YarnClientImpl: Submitted application application_1773395565039_0001
2026-03-13 15:58:56,522 INFO mapreduce.Job: The url to track the job: http://bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:8088/proxy/application_1773395565039_0001/
2026-03-13 15:58:56,522 INFO mapreduce.Job: Running job: job_1773395565039_0001
2026-03-13 15:59:00,559 INFO mapreduce.Job: Job job_1773395565039_0001 running in uber mode : false
2026-03-13 15:59:00,560 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-13 15:59:03,586 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-13 15:59:07,598 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-13 15:59:08,607 INFO mapreduce.Job: Job job_1773395565039_0001 completed successfully
2026-03-13 15:59:08,652 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=144425
                FILE: Number of bytes written=1116570
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1777366
                HDFS: Number of bytes written=16
                HDFS: Number of read operations=11
```

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                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=2234
                Total time spent by all reduces in occupied slots (ms)=1123
                Total time spent by all map tasks (ms)=2234
                Total time spent by all reduce tasks (ms)=1123
                Total vcore-milliseconds taken by all map tasks=2234
                Total vcore-milliseconds taken by all reduce tasks=1123
                Total megabyte-milliseconds taken by all map tasks=2287616
                Total megabyte-milliseconds taken by all reduce tasks=1149952
        Map-Reduce Framework
                Map input records=13130
                Map output records=13129
                Map output bytes=118161
                Map output materialized bytes=144431
                Input split bytes=198
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=144431
                Reduce input records=13129
                Reduce output records=2
                Spilled Records=26258
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=16
                CPU time spent (ms)=2140
                Physical memory (bytes) snapshot=1057992704
                Virtual memory (bytes) snapshot=8367329280
                Total committed heap usage (bytes)=1572864000
                Peak Map Physical memory (bytes)=381526016
                Peak Map Virtual memory (bytes)=2787573760
                Peak Reduce Physical memory (bytes)=295309312
                Peak Reduce Virtual memory (bytes)=2792235008
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=1777168
        File Output Format Counters
                Bytes Written=16
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -cat /avgoutput/part-r-00000
1901    46
1902    21
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$
```

MEANMAX

```
1901    46
1902    21
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -rm -r /meanmaxoutput
rm: `/meanmaxoutput': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar meanmax.MeanMaxDriver /weather /meanmaxoutput
2026-03-13 16:05:30,143 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-13 16:05:30,276 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-13 16:05:30,305 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1773395565039_0002
2026-03-13 16:05:30,453 INFO input.FileInputFormat: Total input files to process : 2
2026-03-13 16:05:30,529 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773395565039_0002
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 16:05:30,682 INFO conf.Configuration: resource-types.xml not found
2026-03-13 16:05:30,682 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-13 16:05:30,709 INFO impl.YarnClientImpl: Submitted application application_1773395565039_0002
2026-03-13 16:05:30,725 INFO mapreduce.Job: The url to track the job: http://bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:8088/proxy/application_1773395565039_0002/
2026-03-13 16:05:30,726 INFO mapreduce.Job: Running job: job_1773395565039_0002
2026-03-13 16:05:34,760 INFO mapreduce.Job: Job job_1773395565039_0002 running in uber mode : false
2026-03-13 16:05:34,761 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-13 16:05:37,786 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-13 16:05:41,800 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-13 16:05:41,806 INFO mapreduce.Job: Job job_1773395565039_0002 completed successfully
2026-03-13 16:05:41,847 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=118167
                FILE: Number of bytes written=1064069
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1777366
                HDFS: Number of bytes written=72
                HDFS: Number of read operations=11
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=2213
                Total time spent by all reduces in occupied slots (ms)=1047
                Total time spent by all map tasks (ms)=2213
                Total time spent by all reduce tasks (ms)=1047
                Total vcore-milliseconds taken by all map tasks=2213
                Total vcore-milliseconds taken by all reduce tasks=1047
                Total megabyte-milliseconds taken by all map tasks=2266112
                Total megabyte-milliseconds taken by all reduce tasks=1072128
        Map-Reduce Framework
                Map input records=13130
                Map output records=13129
                Map output bytes=91903
                Map output materialized bytes=118173
                Input split bytes=198
                Combine input records=0
                Combine output records=0
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                Total time spent by all reduce tasks (ms)=1047
                Total vcore-milliseconds taken by all map tasks=2213
                Total vcore-milliseconds taken by all reduce tasks=1047
                Total megabyte-milliseconds taken by all map tasks=2266112
                Total megabyte-milliseconds taken by all reduce tasks=1072128
        Map-Reduce Framework
                Map input records=13130
                Map output records=13129
                Map output bytes=91903
                Map output materialized bytes=118173
                Input split bytes=198
                Combine input records=0
                Combine output records=0
                Reduce input groups=12
                Reduce shuffle bytes=118173
                Reduce input records=13129
                Reduce output records=12
                Spilled Records=26258
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=15
                CPU time spent (ms)=2030
                Physical memory (bytes) snapshot=1037684736
                Virtual memory (bytes) snapshot=8376963072
                Total committed heap usage (bytes)=1572864000
                Peak Map Physical memory (bytes)=381546496
                Peak Map Virtual memory (bytes)=2792185856
                Peak Reduce Physical memory (bytes)=281702400
                Peak Reduce Virtual memory (bytes)=2799857664
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=1777168
        File Output Format Counters
                Bytes Written=72
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -cat /meanmaxoutput/part-r-00000
01      4
02      1
03      6
04      34
05      89
06      143
07      182
08      172
09      123
10      73
11      21
12      3
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$
```